Research Proposal

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Abstract. EPF has recently seen a resurgence in interest by companies whose operations are energy-dependent, and therefore benefit from timing their energy use in synchrony with the lowest energy prices. However, the literature vastly underreports on the effects of weather data as a key factor for EPF.
The use of multi-feature models that include such data also lags behind the current advancements in Transformer and RNN model architectures, as well as preprocessing opportunities. This research proposes a plan to test multiple architectures and preprocessing techniques in a modular way. We hope to find in its midst, a model that out-performs the current solution used by OpenRemote.

1 Introduction

Energy price forecasting is a highly sought after technique that adds value to companies that benefit from load balancing and management. With the further decarbonisation of the energy grid, renewable energy introduces a strong temporal characteristic to the challenge; one that Neural Networks, particularly advancements in RNN[2, 28, 23, 8, 17], Transformer[13, 26, 20, 27] and ensemble[3, 7, 10] technology can leverage to improve the accuracy of predictions. As feature extraction mechanisms grow more robust[7, 2, 28, 21, 23], and models involve complex sets of features like weather[22, 26], location, and energy consumption[21], the demand for better, more accurace, and more farreaching models is growing withing the industry.

One of the industry players with a keen interest in the technology is OpenRemote. The company provides a platform where entities with multiple energy-demanding components can manage their resources. One of the microservices that OpenRemote provides is an Energy Price Forecasting (EPF) model hosted by Prophet, that does not employ Neural Networks for its predictions[25]. Their current service is only able to offer 24h predictions ahead of time, which causes constraint for schedueled load balancing, and limits the amount of saving a client could make. OpenRemote argues that a 48h prediction window would be more benefitial. Furthemore, the Prophet model is more accurate around the peaks, but relatively innacurate anywhere else. This is due to the limitations of the modeling techniques employed which, although easy to configure and faster than a Neural Network to train and generate forecastings, struggles with the complexities of multi-feature temporal and spacial data that RNNs and Transformers capture with greater ease.

This proposal will explore the literary zeitgeist around Energy-adjacent forecasting, and discuss their methodologies comparatively, while taking into account gaps in the research body. Then, it will discuss the properties that the dataset should seek to include, as to deliver a robust and complete set of features that can later be used to extract more features, or potentially scrapped. Afterwords, it will delimite the research design by mainly outlining the modelling techniques that are going to be used, amongst other procedures. This proposed research should deliver a series of methods that address the challenges of EPF, particularly the request 48h forecasting window, and overall accuracy.

2 Literature Review

One important stream in the literature uses recurrent neural networks and decomposition methods to improve forecasting quality. Fang et al.propose a deep learning framework that combines improved

singular spectrum analysis with a recurrent neural network to support adaptive decomposition and multi-feature fusion for energy consumption forecasting[8]. Although their paper is about energy consumption rather than electricity price, it is relevant because it shows that recurrent models become much stronger when they are combined with preprocessing and feature-fusion techniques rather than used alone. Similarly, another paper combines EEMD, MRMR feature ranking, and BiLSTM, and reports improved forecasting performance in both 1-hour and 24-hour settings[23]. Together, these studies suggest that recurrent architectures still matter, especially when they are paired with strong feature extraction.

This same logic appears in newer electricity-price work. Aksu et al. propose a two-stage day-ahead EPF framework in which the signal is first decomposed with EMD and VMD, after which GRU is used for high-frequency components and LSTM for the remaining components[2]. Their hybrid model reportedly reduces RMSE by about 27.15% compared with traditional machine-learning baselines and by 28.24% compared with LSTM-based baselines. This is important because it suggests that decomposition is not just a preprocessing convenience, but a way to isolate rapid price fluctuations and improve prediction accuracy in volatile markets.

Older work also supports frequency-aware modeling. A 2008 paper propose an adaptive wavelet neural network for electricity-market price forecasting and report better performance than several competing neural and statistical baselines[21]. A more recent paper reinforces preprocessing as an integral step[1]. By combining neural networks with a metaheuristic optimization method, the authors were able to improve training and avoid poor local minima. These papers are not identical to the OpenRemote use case, but they reinforce a broader theme in the literature: wavelet-based decomposition and optimization-enhanced learning are useful when the underlying series is nonlinear, noisy, and unstable.

A second major direction is the use of attention and Transformer-based architectures. Wang et al. present MultiDeT, a Transformer-based method for multi-energy load forecasting in integrated energy systems[26]. Their model uses a one-encoder, multi-decoder structure so that different forecasting tasks can share one representation while learning different task-specific outputs. Even though OpenRemote does not necessarily need multiple decoders, the paper is still relevant because it shows how attention mechanisms can model interactions among multiple related variables. In other words, the core contribution is not only the multi-task architecture itself, but also the idea that one shared encoder can learn a useful multivariate representation.

A more directly relevant paper proposes a modified multi-head Transformer for multivariable building-energy forecasting[20]. The authors compare the Transformer against LSTM and GRU baselines and report a lower MAPE for the Transformer-based model. This is useful for the current project because it supports the idea that attention-based architectures may outperform standard recurrent baselines when multiple input variables need to be combined effectively. The paper also makes an important practical point: LSTM and GRU remain meaningful baselines, but they may not be the best final choice when variable interaction becomes central to the task.

The most relevant recent paper for the OpenRemote use case is Temporal feature mixed inverted transformer: An inverted transformer for effective real-time electricity price forecasting[26]. This study proposes a TFM-iTransformer model that combines an iTransformer backbone with volatility indicators, rate-of-change information, and future covariates. The paper is highly relevant because it explicitly addresses two problems that also apply to OpenRemote: electricity prices are highly volatile, and many forecasting models still underuse known future variables such as weather or dispatch-related information. The reported contribution is therefore not just architectural novelty, but a better way of modeling price uncertainty together with multivariate future information.

Long-horizon forecasting is another recurring concern. Ma and Zhang improve Autoformer by introducing a multi-scale feature fusion network and date-time encoding, producing the MD-Autoformer model[16]. Their experiments show that the modified model improves on Autoformer across several datasets, but they also note that performance becomes difficult at prediction length 192. This is especially significant for the OpenRemote problem, because 48 hours at 15-minute intervals also equals 192 prediction steps. The implication is that long-range forecasting is possi-

ble, but it remains challenging even for advanced Transformer-like models.

Hybrid architectures are also promising. Al-Ali et alċombine CNN, LSTM, and Transformer blocks for solar-energy production forecasting and report that the hybrid outperforms several simpler model combinations[3]. Another paper that employes ensemble models proposes a VMD-based multi-attention feature fusion model for electricity price forecasting that integrates GRU, TCN, SENet, and multi-head attention, and they report strong results against several comparison models[28]. These studies suggest that the best-performing energy-forecasting models increasingly rely on architectural combinations that let different modules specialize in short-term fluctuations, periodic patterns, and feature weighting.

Finally, the broader survey literature supports these patterns. Kong et alȧrgue that strong time-series forecasting models increasingly depend on external variables, multi-horizon design, and better feature extraction methods[13]. Chen et alṡimilarly propose a multivariate forecasting model that explicitly combines temporal features, periodicity, and cross-variable spatial relationships through LSTM, attention, graph-based periodic modeling, and CNNs[7]. Manero et al; in a literature review on wind forecasting, also conclude that CNN and RNN/LSTM approaches outperform simple MLP approaches, especially when exogenous environmental variables are available[17]. Across these sources, the same conclusion appears repeatedly: energy forecasting improves when models can exploit multiple correlated inputs rather than treating the target series in isolation.

However, there is a considerable gap on the explicit inclusion of weather data for EPFȦ 2016 survey on Wind Energy, Load and Price forecasting mentions weather data as '[playing] a vital role in load forecasting', yet their approach has a higher degree of codependency between variables[18]. Weather patters are more linearly correlated to energy load than prices (see Fig.1). Sgarlato et alṡtrongly show that the inclusion of weather predictions has strong explanatory power[22]. This finding is key for our mission of expanding the forecasting window to 48h. However, it is worth noting that this correlation is tied to location. Other papers discuss the use of weather data for energy forecasting[4, 6]. Despite these papers supporting the inclusion of weather data, they use older methodology, or simply limit themselves to deterministic models.

Lastly, no paper included xLSTM[5] in their methodology, as expected when such is a recent technology. Despite its immediate use being for NLP, it has been used sucesfully for time-series predictions[11, 14]. Its use could limit computational power while conserving the benefits of attention used in Transformers.

3 Methodology

3.1 Data

The dataset must comprehend be as detailed as possible. The ideal $\vec{\Delta}$ would be 15 minutes, but 1 hour would also be functional. It remains to be seen if a larger $\vec{\Delta}$ of 1 hour with statistical summaries is benefitial, and it might be worth testing.

The dataset should include the price and energy production/consumption for the the netherlands. Weather data for the Netherlands would be ideal. We could cluster different sensor points, and use location as a feature, however we should start simpler, by using a national average. Temporal features should also be included, such as weekend, or holiday flag. All features should be interpolated on timestamp.

This dataset would be good enough for modelling.

3.2 Research Design

This section will outline the methodologies to be experiemented with. These might encompass feature extraction, data processing, model architecture, or all. We will, for the purposes of this proposal, use a naming convention of Exp.$i.v$, where i is the number of the experiment configuration, and v is a letter, whose order is reset for each i.

3.2.1 Exp.1

From our literature, one of the most recurrent topics is the power of ensemble models, particularly ones using RNNs. To start out simple, we will introduce a CNN-LSTM variant architecture as a base and design multiple experiments that build upon it.

Since we are using CNNs, which serve the purpose of capturing long-term sequential patters, it would be ideal to test an array of pre-

	Exp.1			
	a	b	c	d
Architecture	CNN-BiLSTM	CNN-xLSTM	CNN-BiLSTM-Transformer	CNN-Transformer
Preprocessing methods	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST
Additional notes				To prove null-hypothesis i.e. RNNs punish accuracy

Table 1: Exp.1: Model Architectures and Preprocessing

processing methods, some of which enhance 3-dimentional features. For Exp.1, all variations will cycle through:

1. Normalisation
2. Wavelet Transform[21, 9]
3. Transformer Encoding (PatchTST[19])

Exp.1.a Variation **a** will use a simple CNN-BiLSTM, since the use of BiLSTM consistently outperforms normal LSTMs in time-series forecasts.

Exp.1.b Variation **b** will use a CNN-xLSTM.

Exp.1.c Variation **c** will use a CNN-BiLSTM-Transformer. The discrepancy between results for each preprocessing method should be greatly scrutinised for this variation, as the use of a Transformer encoder and output model essentially places the CNN and BiLSTM as feature extractors.

Exp.1.d Variation **d** will use a CNN-Transformer. By removing the transformer, we aim to prove the null hypothesis that RNNs, particularly LSTMs, are greatly outperformed by Transformers for short-term dependencies, and that the features that they extract act as noise for a transformer.

This set of variations offer a comprehensive use of the methods that the literature has provided, but they also take into account standard practises i.e. Normalisation and the use of CNNs for time-series data, also found in the literature[3, 28, 17].

3.2.2 Exp.2

Current trends in AI strongly favour Transformer architectures, or Attention layers. Exp.2 will focus on exploring this particular aspect in more detail. Exp.1 also features Transformers, but in this section, we want to amplify their strenghts by means of expanding our number of features. Non-linearity is accutely endemic to all of our features[17]. One hypothesis is that the Transformer architecture will be able to capture linear features where there is usually non-linearity.

Exp.2.a Variation **a** will use a classic multi-head Decoder-Only transformer[20]. Decoder-Only transformers perform well in renewable scenario analysis, and use less memory[10]. This is relevant for our study because the nature of the data and analysis is similar by virtue of the non-linear and complex time-series tasks both try to address.

Exp.2.b Variation **b** will use the same model as Exp.2.a, but utilise Fast Fourier Transformers for feature extraction[29].

Exp.2.c Variation **c** will use a multi-head Encoder-Decoder Transformer. Encoder-Decoder variants perform great with forecasting tasks[10].

Exp.2.d Variation **d** will use a Kernel-based Transformer. This architecture can better capture linearity when it exists[24] which has been pointed out as a reason for transformers failing to generalise on Time-Series tasks[12] , all the while using less memory.

Exp.2.e Variation **e** will use an inverted Transformer, henceforth known as iTransformer; a method introduced by a landmark paper back in 2023[15]. The paper specifically demonstrates the effectiveness of the proposed methodology for Time-Series Forecasting, and has been explicitely used for EPF[26].

	Exp.2				
	a	b	c	d	e
Architecture	Decoder-Only Transformer	Decoder-Only + FFT	Encoder-Decoder Transformer	Kernel-based Transformer	iTransformer
Preprocessing methods	Standard	Fast Fourier Transform	Standard	Kernel-based	Standard
Additional notes		Same model as **a**			

Table 2: Exp.2: Model Architectures and Preprocessing

4 Preliminary Data

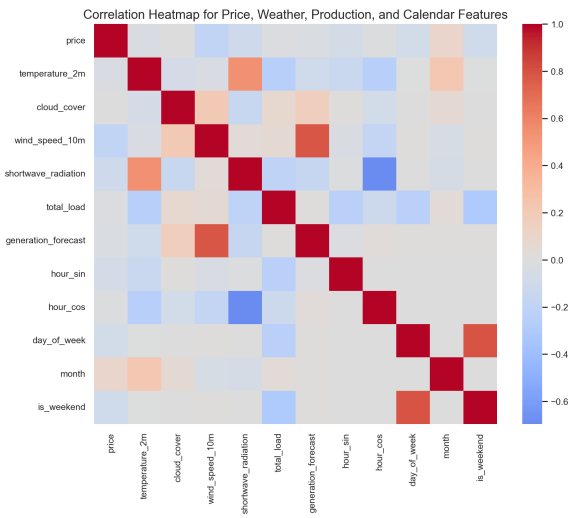

Figure 1: Heatmap of correlation between features

A preliminary analysis of the features in our dataset confirms some of the presumptions gathered from the literature; that price is inversely correlated to wind speed, with shortwave radiation also being slightly inversely correlated, as seen on Fig.1. Its rather curious to observe the week correlation between load and price.

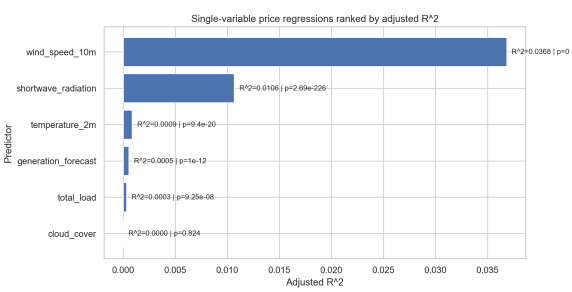

Figure 2: Statistical significance per feature

Similarly, wind speed, solar radiance, temperature, energy generation and load all appear to be statistically significant towards price (Fig.2).

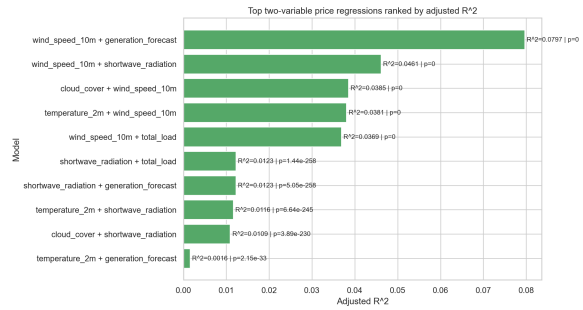

Figure 3: Statistical significance per feature pair

But interestingly, the combination of wind speed and overall generation seem to be more statistically significant, and hold greater explanatory power, than a combination of wind and solar. This interpretation is supported by Fig.1, which shows that wind speed and generation forecast are closely correlated, meaning that wind power makes up a sizable chunk of the energy produced, when the conditions allow for it. Besides, combinations of different weather features are all statistically significant.

5 Statement of Limitations

This research is limited by multiple factors, such as time that the members can put into the development of the research, as well as the time-window that it needs to be done. Furthermore, the experience of the members of this research group is being actively built on this very project. Therefore, its execution may contain flaws in methodology. Hardware limitations are also to be considered, as complex models require better hardware to be properly tested, with the biggest data input as it could be otherwise. This research will attempt to rank the proposed methods in an attempt to find a good match for EPF, and help the students working on the project learn and practice the scope of the subject.

6 Conclusion

EPF stands to be one of the most substantial uses of Artificial Intelligence for companies who's energy used can be timed, and therefore optimised. This research will contribute to the validation of existing methods within the literature, by expanding on their use case and exposing them to a new set of features. We will also test methods used in other domains, proposing that adapting them to EPF might be beneficial. Lastly, by comparing similar methods in a methodical and modular way, we may draw conclusions on what methods in particular better or worsen the performance of a particular model. By doing so, we may find a model architecture that delivers on the accuracy and lag-prediction requested by OpenRemote.

References

[1] Oveis Abedinia, Nima Amjady, and Noradin Ghadimi. Solar energy forecasting based on hybrid neural network and improved metaheuristic algorithm. *Computational Intelligence*, 34(1), October 2017.

[2] İnayet Özge Aksu, Sina Ghaemi, and Amjad Anvari-Moghaddam. A two-stage deep learning approach for accurate day-ahead electricity price forecasting. *Engineering Applications of Artificial Intelligence*, 163:112721, January 2026.

[3] Elham M. Al-Ali, Yassine Hajji, Yahia Said, Manel Hleili, Amal M. Alanzi, Ali H. Laatar, and Mohamed Atri. Solar energy production forecasting based on a hybrid cnn-lstm-transformer model. *Mathematics*, 11(3), 2023.

[4] José R. Andrade and Ricardo J. Bessa. Improving renewable energy forecasting with a grid of numerical weather predictions. *IEEE Transactions on Sustainable Energy*, 8(4):1571–1580, 2017.

[5] Maximilian Beck, Korbinian Pöppel, Markus Spanring, Andreas Auer, Oleksandra Prudnikova, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. xlstm: Extended long short-term memory, 2024.

[6] Kristijan Brecl and Marko Topič. Photovoltaics (PV) system energy forecast on the basis of the local weather forecast: Problems, uncertainties and solutions. *Energies*, 11(5):1143, May 2018.

[7] Junfeng Chen, Azhu Guan, Jingjing Du, and Altangerel Ayush. Multivariate time series prediction with multi-feature analysis. *Expert Systems with Applications*, 268:126302, 2025.

[8] Lei Fang and Bin He. A deep learning framework using multi-feature fusion recurrent neural networks for energy consumption forecasting. *Applied Energy*, 348:121563, October 2023.

[9] Komei Hiruta and Junsuke Senoguchi. Wavelet-cnn for temporal data: Enhancing long-term stock price prediction via multi-resolution wavelet decomposition and cnn-based feature extraction. *International Journal of Information Management Data Insights*, 5(2):100360, December 2025.

[10] Umair Hussan, Huaizhi Wang, Jianchun Peng, Hui Jiang, and Hamna Rasheed. Transformer-based renewable energy forecasting: A comprehensive review. *Renew. Sustain. Energy Rev.*, 226(116356):116356, January 2026.

[11] Lei Kang, Xuanshuo Fu, Javier Vazquez-Corral, Ernest Valveny, and Dimosthenis Karatzas. xlstm-ecg: Multi-label ecg classification via feature fusion with xlstm, 2025.

[12] Yekun Ke, Yingyu Liang, Zhenmei Shi, Zhao Song, and Chiwun Yang. Curse of attention: A kernel-based perspective for why transformers fail to generalize on time series forecasting and beyond, 2024.

[13] Xiangjie Kong, Zhenghao Chen, Weiyao Liu, Kaili Ning, Lechao Zhang, Syauqie Muhammad Marier, Yichen Liu, Yuhao Chen, and Feng Xia. Deep learning for time series forecasting: a survey. *International Journal of Machine Learning and Cybernetics*, 16(7–8):5079–5112, February 2025.

[14] Maurice Kraus, Felix Divo, Devendra Singh Dhami, and Kristian Kersting. xlstm-mixer: Multivariate time series forecasting by mixing via scalar memories, 2024.

[15] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting, 2023.

[16] Xiangkai Ma and Huaxiong Zhang. Time series forecasting method based on multi-scale feature fusion and autoformer. *Applied Sciences*, 15(7):3768, March 2025.

[17] Jaume Manero, Javier Béjar, and Ulises Cortés. Wind energy forecasting with neural networks: A literature review. *Computación y Sistemas*, 22(4), December 2018.

[18] Y. Nagaraja, T. Devaraju, M. Vijay Kumar, and Sreedhar Madichetty. A survey on wind energy, load and price forecasting: (forecasting methods). In *2016 International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT)*, pages 783–788, 2016.

[19] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers, 2022.

[20] Hugo S. Oliveira and Helder P. Oliveira. Transformers for energy forecast. *Sensors*, 23(15):6840, August 2023.

[21] N. M. Pindoriya, S. N. Singh, and S. K. Singh. An adaptive wavelet neural network-based energy price forecasting in electricity markets. *IEEE Transactions on Power Systems*, 23(3):1423–1432, 2008.

[22] Raffaele Sgarlato and Florian Ziel. The role of weather predictions in electricity price forecasting beyond the day-ahead horizon. *IEEE Transactions on Power Systems*, 38(3):2500–2511, 2023.

[23] Zhen Shao, Qingru Zheng, Chen Liu, Shuangyan Gao, Gang Wang, and Yan Chu. A feature extraction- and ranking-based framework for electricity spot price forecasting using a hybrid deep neural network. *Electric Power Systems Research*, 200:107453, November 2021.

[24] Yu-Chuan Su and Kristen Grauman. Kernel transformer networks for compact spherical convolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9442–9451, 2019.

[25] Sean J Taylor and Benjamin Letham. Forecasting at scale. September 2017.

[26] Baichun Wang, Baoxian Huang, Qinglun Zhang, Yan Shi, and Hong Men. Temporal feature mixed inverted transformer: An inverted transformer for effective real-time electricity price forecasting. *Engineering Applications of Artificial Intelligence*, 167:113778, March 2026.

[27] Chen Wang, Ying Wang, Zhetong Ding, Tao Zheng, Jiangyi Hu, and Kaifeng Zhang. A transformer-based method of multienergy load forecasting in integrated energy system. *IEEE Transactions on Smart Grid*, 13(4):2703–2714, 2022.

[28] Yuzhen Xu, Xin Huang, Ziao Gao, Mohamed A. Mohamed, and Tao Jin. A novel electricity price forecasting approach based on multi-attention feature fusion model optimized by variational mode decomposition. *Measurement*, 253:117596, September 2025.

[29] Kyrylo Yemets, Ivan Izonin, and Ivanna Dronyuk. Time series forecasting model based on the adapted transformer neural network and fft-based features extraction. *Sensors*, 25(3):652, January 2025.